

Rosenberg mathematical routes

Scope

Six mathematical routes through the 33 approved Rosenberg Cards. Shared Cards are reused across routes and count once in the Catalog. The source-only density extension is separate.

Mathematical routes

Representations, irreducibility and equivalence

Representations without a unit equation, nonzero irreducibility and unitary equivalence set the language of the theorem. The singleton condition names one representative of the unique nonzero irreducible class.

4 direct Cards + 0 reused prerequisites = 4 unique Cards. Shared Cards count once in this route.

- Key definition: [A representative of the unique irreducible class](#)
- Key definition: [Representations without a unit-preservation requirement](#)

[Read this route with prerequisites](#)

Pure states and faithfulness

A pure state detects a hypothetical nonzero kernel element and its GNS representation is irreducible. Unitary equivalence to the singleton representative then contradicts that kernel element, giving faithfulness and the related simplicity consequence.

5 direct Cards + 5 reused prerequisites = 10 unique Cards. Shared Cards count once in this route.

- Key result: [Faithfulness from a unique irreducible class](#)
- Key result: [A faithful singleton model forces simplicity](#)

[Read this route with prerequisites](#)

Characters, eigenvectors and a scalar corner

Characters of a maximal abelian subalgebra of the unitization extend to pure states. A character distinct from the scalar character yields a joint unit eigenvector in the singleton model. Countability and an isolated non-scalar character then give a nonzero scalar-corner projection and its rank-one image.

11 direct Cards + 8 reused prerequisites = 19 unique Cards. Shared Cards count once in this route.

Every compact operator has a preimage

One represented rank-one projection yields every rank-one operator using dense orbits. Closed-range and compact approximation arguments put all compact operators in the image of the original algebra.

3 direct Cards + 16 reused prerequisites = 19 unique Cards. Shared Cards count once in this route.

- Key result: [Compact operators lie in a singleton representation range](#)

- Key result: [A character becomes a joint unit eigenvector](#)
- Key result: [A projection represented by a rank-one operator](#)

[Read this route with prerequisites](#)

- Key result: [Every rank-one operator has an algebra preimage](#)

[Read this route with prerequisites](#)

Every represented operator is compact

The preimage ideal of compact operators and a separating character exclude a noncompact represented image. The contradiction uses the same projection while forcing the character to take both zero and one.

7 direct Cards + 14 reused prerequisites = 21 unique Cards. Shared Cards count once in this route.

- Key result: [Every operator in a singleton image is compact](#)
- Key definition: [The preimage ideal of the compact operators](#)

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Compact-operator models and unital consequences

Faithfulness and both compact-image inclusions identify the nonunital compact-operator model in an unbundled sense. In the unital singleton case the identity becomes compact, yielding finite dimension and the exact full-operator image consequences.

12 direct Cards + 20 reused prerequisites = 32 unique Cards. Shared Cards count once in this route.

- Key result: [A singleton model is faithful and exactly compact-valued](#)
- Unital consequence: [Finite-dimensional algebra in the unital singleton case](#)

[Read this route with prerequisites](#)